

Regional diversification of highly pathogenic avian influenza clade 2.3.4.4b genotypes in Asia and development of an automated genotyping system, GenoFlu-Asia.

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INTRODUCTION

- Clade 2.3.4.4b H5 HPAI viruses have undergone extensive global spread and reassortment, generating diverse genomic constellations.
- Genotype classification is therefore essential for tracking the emergence, persistence, and transmission of reassortant viruses.
- Genotype frameworks have been developed for North America (GenoFLU), Europe (GENIN2), and globally (ggFLU).
- However, genotype definitions in Asia have largely been proposed at the individual-country level, without unified framework integrating regional genomic diversity.
- This limitation hinders comprehensive epidemiological analysis, including characterization of genotype variation across countries and seasons, as well as viral transmission dynamics.
- In this study, we established a standardized classification and nomenclature system for Asian clade 2.3.4.4b H5N1 viruses.
- We also developed GenoFLU-Asia, an automated genotype-assignment tool, and characterized the spatiotemporal and host-associated dynamics of these viruses across Asia.

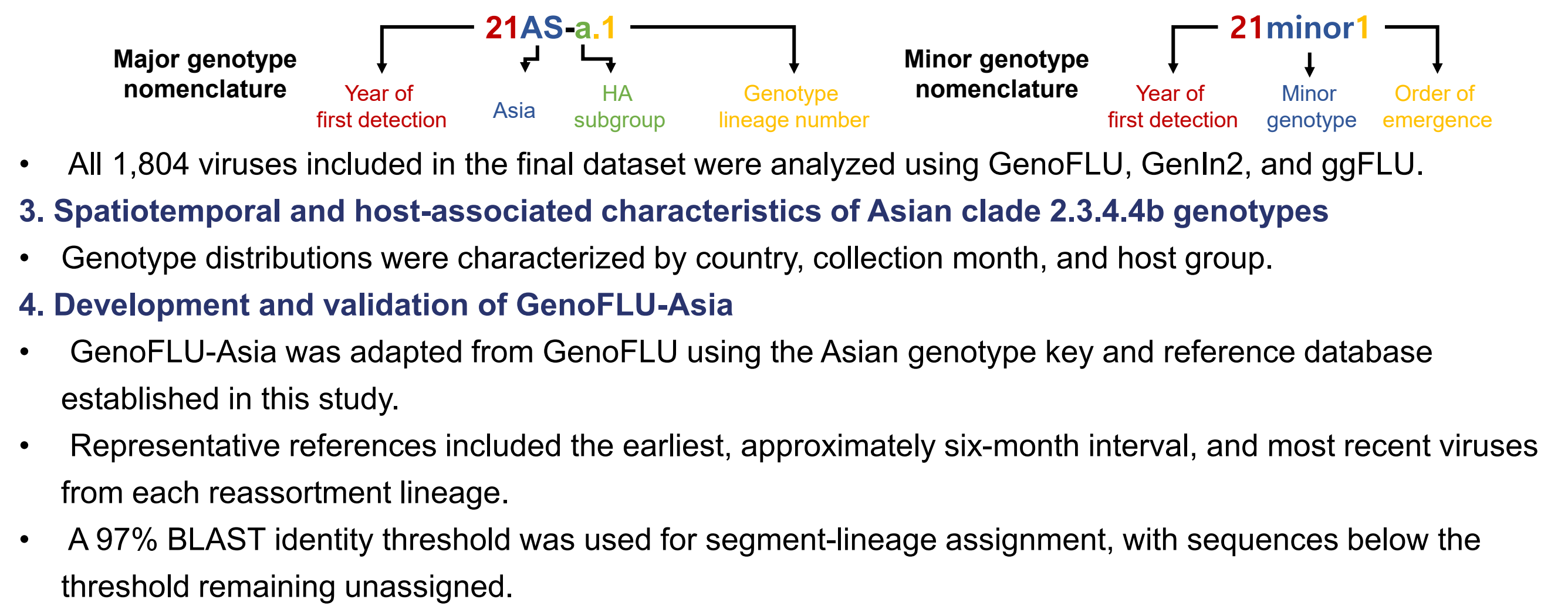
MATERIALS & METHODS

1. Sequence data collection and quality control

- Complete genomes of Asian clade 2.3.4.4b HPAI H5 viruses (2021–2026) and LPAI (2016–2026) reference viruses were retrieved from GISAID.
- Sequences were aligned using MAFFT and filtered by segment-specific minimum length and <5% ambiguous nucleotides.
- In case of clade 2.3.4.4b, only viruses with all eight genomic segments were retained.

2. Genotype assignment through phylogenetic analysis

- Segment-specific approximately ML trees were constructed using FastTree.
- Reassortment lineages were primarily defined by clustering with LPAI references or, when references were unavailable, by $\leq 97\%$ pairwise nucleotide identity between phylogenetically distinct clusters.
- Preliminary genotypes were defined from eight-segment constellations and were validated with ML tree using RAxML (GTR + Gamma; 500 bootstraps).
- Reassortment clusters with $\geq 70\%$ bootstrap support were retained for final genotype definition.
- The workflow for genotype definition of Asian clade 2.3.4.4b viruses is shown in **Figure 1**.
- Genotype nomenclature was established based on the HA subgroups G2a, G2b, G2c, and G2d.
- Genotypes represented by <3 viruses were classified as minor genotypes.



RESULT & DISCUSSION

1. Genotype diversity of Asian clade 2.3.4.4b viruses

- A total of 129 genotypes were identified, comprising 45 major and 84 minor genotypes.
- Genotypes introduced from Europe or North America were identifiable by existing regional systems, including 23AS-a.5(N5) (GENIN2 EA-2021-I; GenoFLU A6) and 22NA.1 (GenoFLU B3.2).
- In comparison with ggFlu, multiple phylogenetically distinct Asian genotypes were assigned to the same global genotype (**Table 1**).
- G2c showed the greatest genotype diversity (21 major and 42 minor genotypes), whereas G2e showed the lowest diversity (3 minor genotypes).
- Novel genotype emergence peaked during the 2022–2023 epidemic season (19 major and 28 minor genotypes) (**Figure 2**).

2. Geographic distribution of Asian genotypes

- Genotype composition was strongly geographically structured: 21eAS-d.1 predominated in Japan, 22AS-c.2 in South Korea, 23AS-c.2.2 in Bangladesh, 21AS-b.1 in Vietnam, and 21AS-a.1(N6) in China (**Figure 3**).

3. Temporal dynamics of Asian genotypes

- In Japan, 21AS-d.1 persisted from the 2021–2022 through the 2025–2026 epidemic season, with substantial co-circulation of 22AS-c.2/2.1.1.1 during 2022–2023.
- South Korea showed sequential genotype turnover from 21AS-b.1/1.3 to 22AS-c.2 (2022–2023), followed by 24AS-c.4/21AS-d.1 co-circulation (2023–2024) and 24AS-c.3.1 predominance (2024–2025).
- Vietnam showed sequential genotype shifts from 21AS-b.1/1.2 during the late 2021–2022 and early 2022–2023 epidemic seasons to 22AS-c.2 in June–September 2023, followed by 22AS-b.1.5 predominance (**Figure 4a**).

4. Host distribution of Asian genotypes

- In Japan, 22AS-c.2.1.1.1 was the second most common genotype among other avian species but was detected in only two poultry viruses, whereas in South Korea, 21AS-b.1.3 was more frequent in poultry (11.17%) than in waterfowl (1.80%).
- Mammalian detections were limited to Japan, South Korea, and China and involved only a subset of the locally circulating genotypes (**Figure 4b**).

5. Discussion

- The predominance of regionally derived genotypes suggests that Asia supports sustained regional circulation, reassortment, and continued diversification of clade 2.3.4.4b viruses.
- The prolonged persistence of genotypes such as 21AS-d.1 may reflect greater compatibility with wild-bird hosts compared with more transient genotypes.
- Despite their geographic proximity and connectivity through wild-bird movements, South Korea and Japan showed distinct patterns of predominant genotype circulation.
- Differences in genotype composition among host groups suggest host-associated circulation; however, the relative contributions of individual host populations to genotype emergence and transmission require further investigation.
- The extensive regional genotype diversity, limited resolution of existing genotyping systems for Asian reassortants, and cross-border circulation of shared genotypes highlight the need for a unified Asia-wide genotype framework.
- GenoFLU-Asia provides a platform for rapid and harmonized genotyping genomic surveillance of clade 2.3.4.4b viruses across Asia.

FIGURES

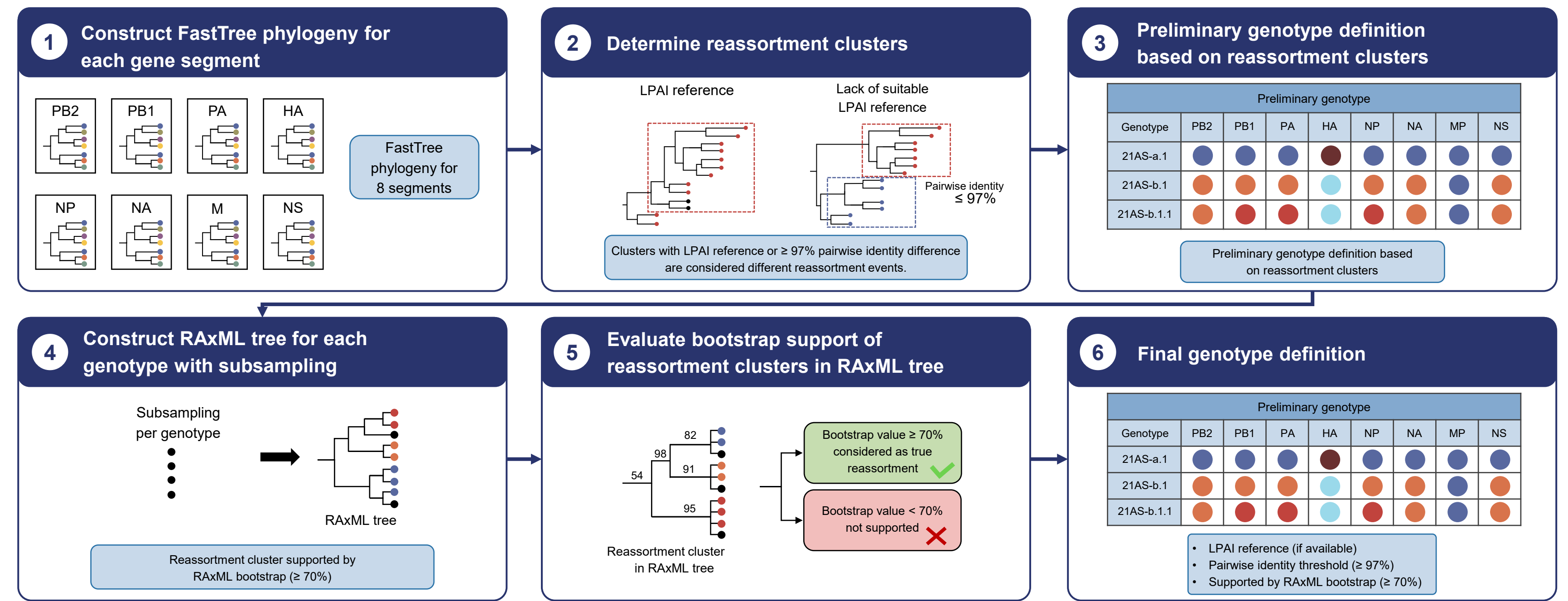


Figure 1. Workflow for genotype definition of Asian clade 2.3.4.4b HPAI H5 viruses. Schematic overview of the stepwise process used to identify reassortment lineages, define preliminary genotypes, and establish final genotype assignments through phylogenetic validation.

HA subgroup	Asia Genotype	Number of sequences	Genotyping tool	HA subgroup	Asia Genotype	Number of sequences	Genotyping tool
			GENIN2				GENIN2
			ggFLU				ggFLU
G2a	21AS-a.1(N6)	28	Unassigned	G2c	22AS-c.2	288	Unassigned
	21AS-a.2(N6)	7	Unassigned		22AS-c.2.1	200	Unassigned
	21AS-a.3(N6)	4	Unassigned		22AS-c.2.1.1	149	Unassigned
	21AS-a.4(N6)	3	Unassigned		22AS-c.2.1.1.1	20	Unassigned
	21AS-a.5(N5)	34	Unassigned		22AS-c.2.1.1.3	14	Unassigned
	21AS-a.6(N6)	4	Unassigned		22AS-c.2.1.1.4	3	Unassigned
G2b	21AS-b.1	191	Unassigned	G2d	23AS-c.2.2	79	Unassigned
	21AS-b.1.1	5	Unassigned		24AS-c.2.2.1	4	Unassigned
	21AS-b.1.2	54	Unassigned		24AS-c.2.2.2	3	Unassigned
	21AS-b.1.3	35	Unassigned		24AS-c.3	4	Unassigned
	22AS-b.1.4	30	Unassigned		24AS-c.3.1	69	Unassigned
	22AS-b.1.4.1	5	Unassigned		24AS-c.3.1.1	9	Unassigned
G2c	22AS-b.1.5	40	Unassigned	NA	25AS-c.3.1.2	9	Unassigned
	23AS-b.1.5.1	3	Unassigned		23AS-c.4	46	Unassigned
	23AS-b.1.5.2	9	Unassigned		21AS-d.1	426	Unassigned
	23AS-b.1.6	3	Unassigned		22AS-d.1.1	5	Unassigned
	21AS-c.1	10	Unassigned		22AS-d.1.2(N2)	3	Unassigned
	21AS-c.1.1	3	Unassigned		22AS-d.1.3	10	Unassigned

Table 1. Comparison of major Asian clade 2.3.4.4b HPAI H5 genotypes with existing genotyping systems. Major genotypes defined in this study are grouped by HA subgroup, with the number of sequences representing each genotype shown. Corresponding genotype assignments from GenoFLU, GENIN2, and ggFlu are provided for comparison.

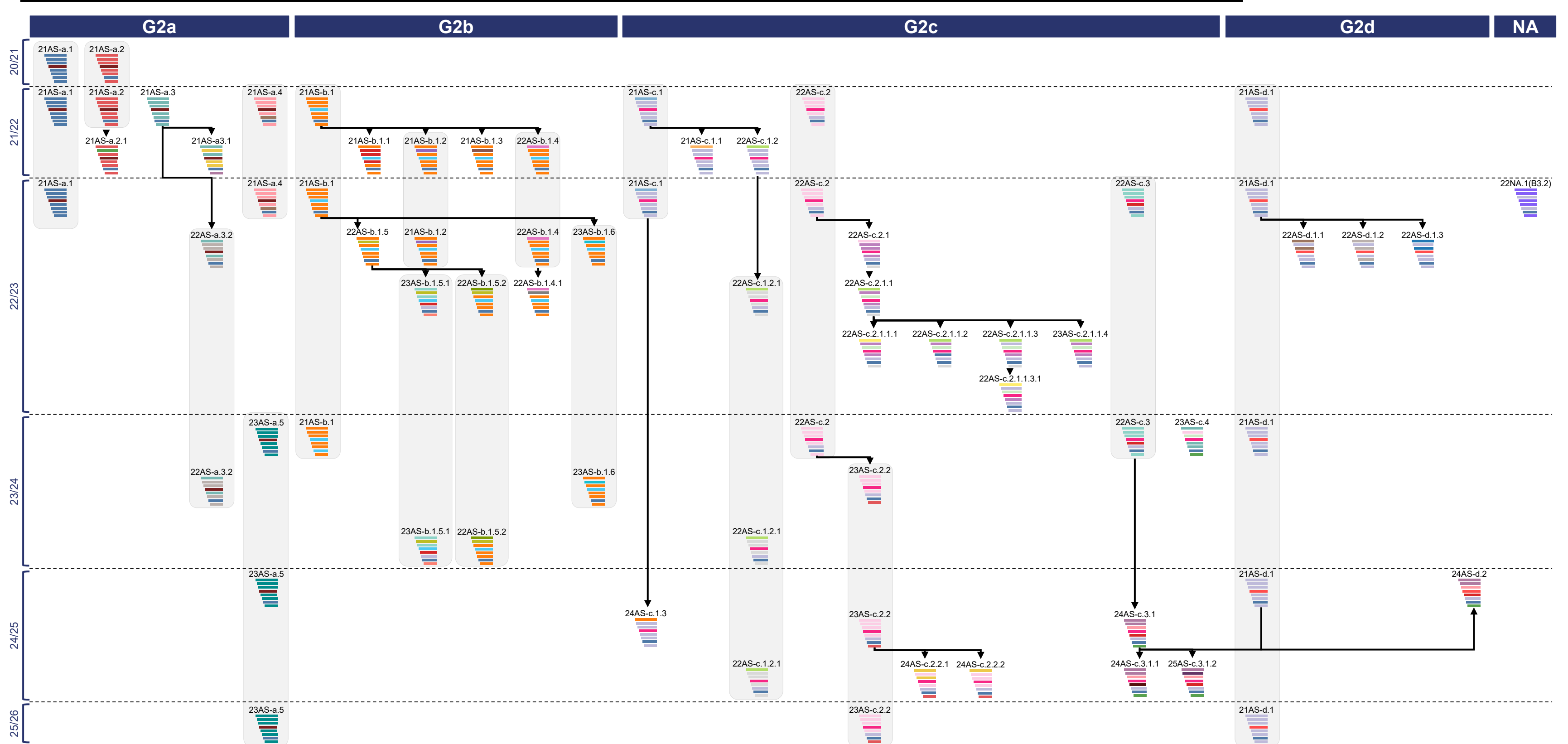


Figure 2. Reassortment patterns and temporal emergence of major Asian clade 2.3.4.4b HPAI H5 genotypes. Major genotypes are grouped according to HA subgroup (G2a–G2d and NA) and arranged by epidemic season of detection. The eight horizontal bars represent the viral genome segments, with different colors indicating distinct reassortment lineages. Arrows indicate inferred descendant genotypes derived from preceding genotype backbones through reassortment. Gray shaded boxes indicate genotypes detected across more than two epidemic seasons.

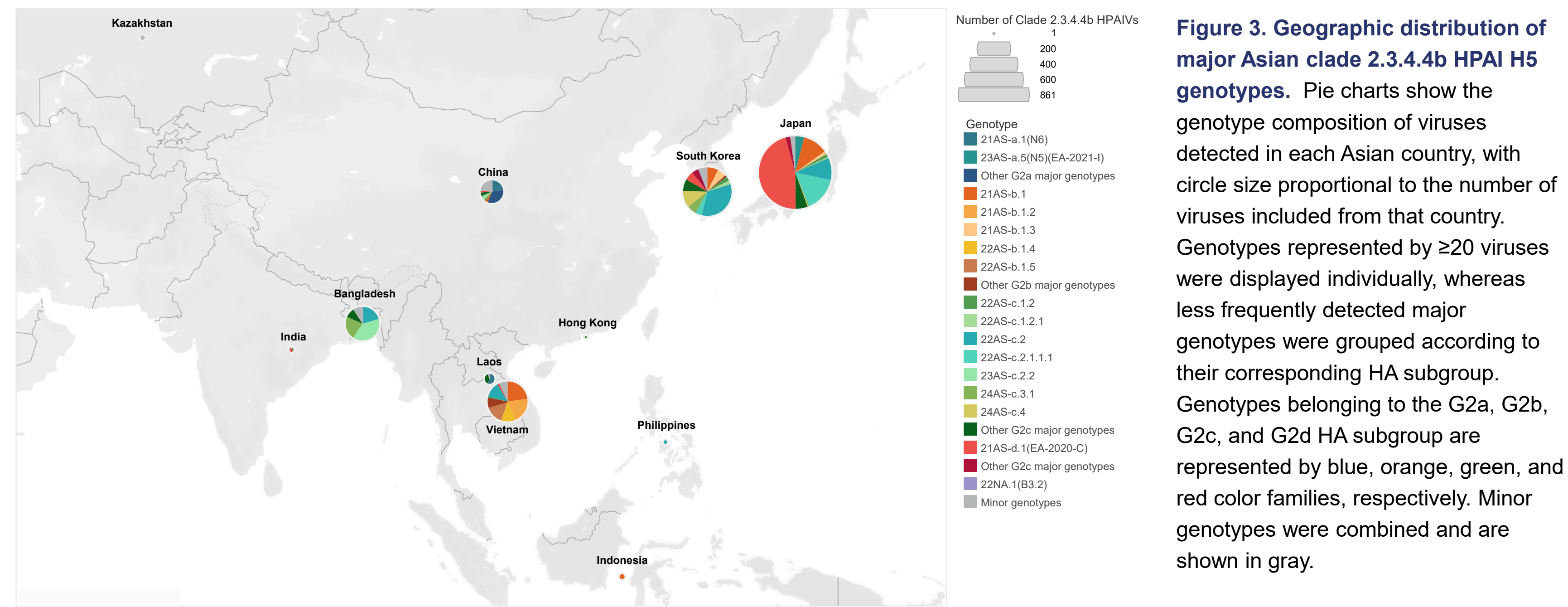


Figure 3. Geographic distribution of major Asian clade 2.3.4.4b HPAI H5 genotypes. Pie charts show the genotype composition of viruses detected in each Asian country, with circle size proportional to the number of viruses included from that country. Genotypes represented by ≥ 20 viruses were displayed individually, whereas less frequently detected major genotypes were grouped according to their corresponding HA subgroup. Genotypes belonging to the G2a, G2b, G2c, and G2d HA subgroup are represented by blue, orange, green, and red color families, respectively. Minor genotypes were combined and are shown in gray.

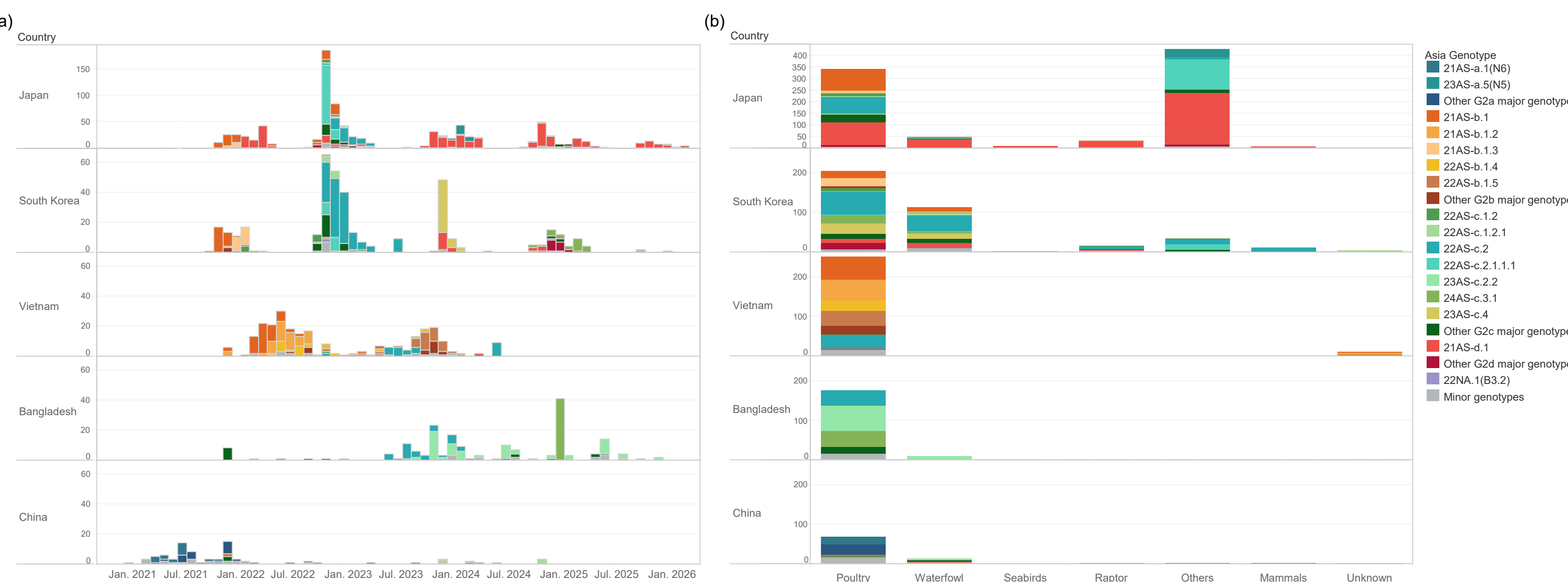


Figure 4. Temporal and host-associated distribution of Asian clade 2.3.4.4b HPAI H5 genotypes. The five countries with ≥ 50 viruses in the study dataset—Japan, South Korea, Vietnam, Bangladesh, and China—are shown. (a) Monthly distribution of genotypes by country throughout the study period. (b) Genotype distribution by host category, classified as poultry, waterfowl, seabirds, raptors, other avian species, mammals, and unknown. Genotypes represented by ≥ 20 viruses were displayed individually, whereas less frequently detected major genotypes were grouped according to their corresponding HA subgroup. G2a, G2b, G2c, and G2d genotypes are represented by blue, orange, green, and red color families, respectively. Minor genotypes were combined and are shown in gray.